HW3

Caution & AI Usage

Please do not import packages (like scikit-learn) that are not listed in the provided code.

For this homework, you are welcome to use AI (e.g., ChatGPT) to understand the models, algorithms, PyTorch, and the code.

However, you are not allowed to directly ask AI to implement the solutions and simply copy and paste them.

Preparation

Please make sure you have gone through:

- Week 2 materials
- Slide decks of Lectures 8 – 11
- Please read Textbook chapters 9 – 14, 24, and 26, at least the portions covered by the lectures.

The notation in this slide deck may differ slightly from the lecture slides! The goal is to make implementation easier.

Preparation

In this homework, you will need to use PyTorch. You may find tutorials here:

https://docs.pytorch.org/tutorials/beginner/basics/intro.html

Topic

In this homework, you are going to experience convolutional neural networks (CNNs) and vision transformers, in particular:

- ResNet-50
- DINO-v2

We demoed both models in Week 2 of this semester. You are also encouraged to check the corresponding papers and blog posts:

- ResNet: paper, slides
- DINO-v2: https://dinov2.metademolab.com/

Google Colab

https://colab.research.google.com/drive/1Sy9nqN8vqeNoNH44xJKXgRogqLhUPRqM?usp=sharing

Please save your own copy and implement there

Part 1: linear probing

As mentioned in class, a neural network classifier can be represented by:

$$\text{Prediction} = \underset{c}{\text{argmax}}\, \boldsymbol{w}_c^T\, f_{\boldsymbol{\theta}}(\boldsymbol{x})$$

- Specifically, $\boldsymbol{x}$ is the input image, $f_{\boldsymbol{\theta}}(\boldsymbol{x})$ is the feature vector extracted by a neural network; $\boldsymbol{w}_c$ is the linear classifier (class profile) of class c

Part 1: linear probing

In Part 1, we will load the feature extractor f_{θ} directly from pre-trained checkpoints, and we will freeze it without updating.

Your goal is to learn $\{\boldsymbol{w}_c\}_{c=1}^{K}$ for *K* categories, such that the classifier will classify test images well.

Part 1: linear probing

Dataset: Caltech-UCSD Birds-200-2011 (CUB-200-2011) with 200 bird species.
https://www.vision.caltech.edu/datasets/cub_200_2011/



Part 1: linear probing

Your **Jobs 1 to 4** include:

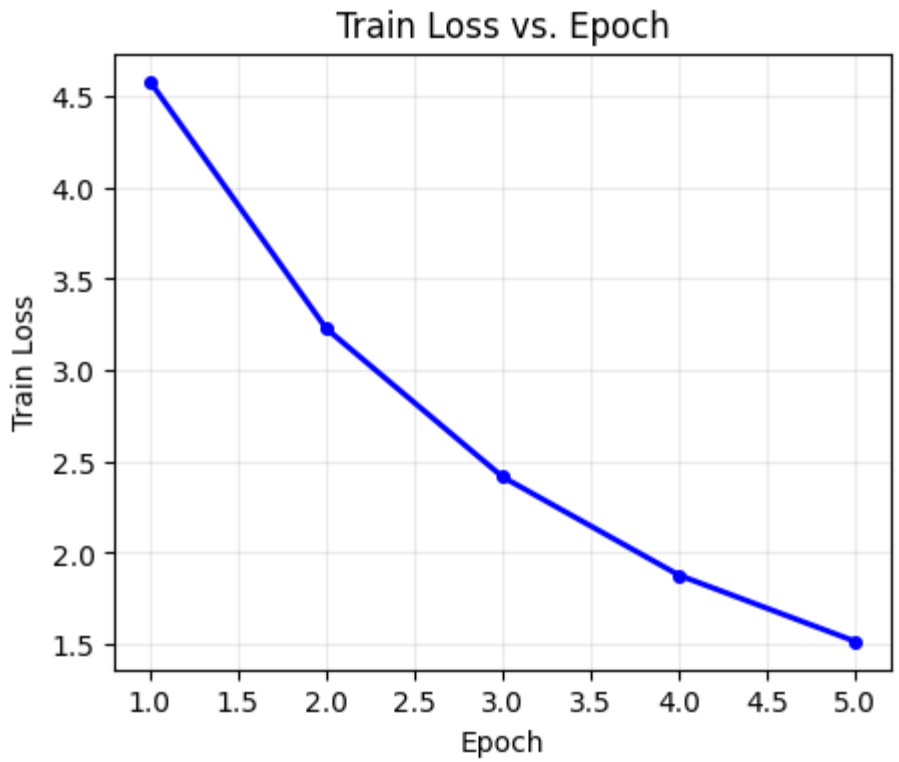
- Set up the loss function (criterion) and optimizer
- Set up the evaluation loop
- Set up gradient computation and model weight update
- Plot the training loss and test accuracy

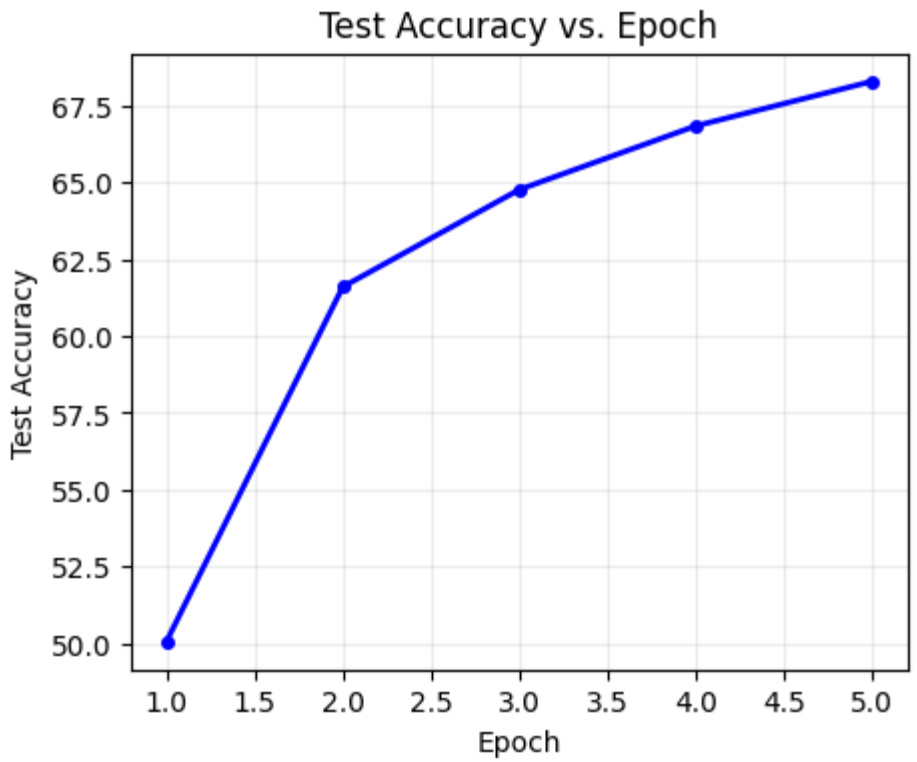
Algorithm 10.1: Gradient descent (GD). Optimizing a cost function $J : \theta \to \mathbb{R}$ by descending the gradient $\nabla_\theta J$.

Input: objective function J, initial parameter vector θ^0, learning rate η, number of steps K
Output: trained parameter vector $\theta^* = \theta^K$
for $k = 0, \ldots, K-1$ **do**
    $\theta^{k+1} \leftarrow \theta^k - \eta \nabla_\theta J(\theta^k)$

Part 1: linear probing

Example results





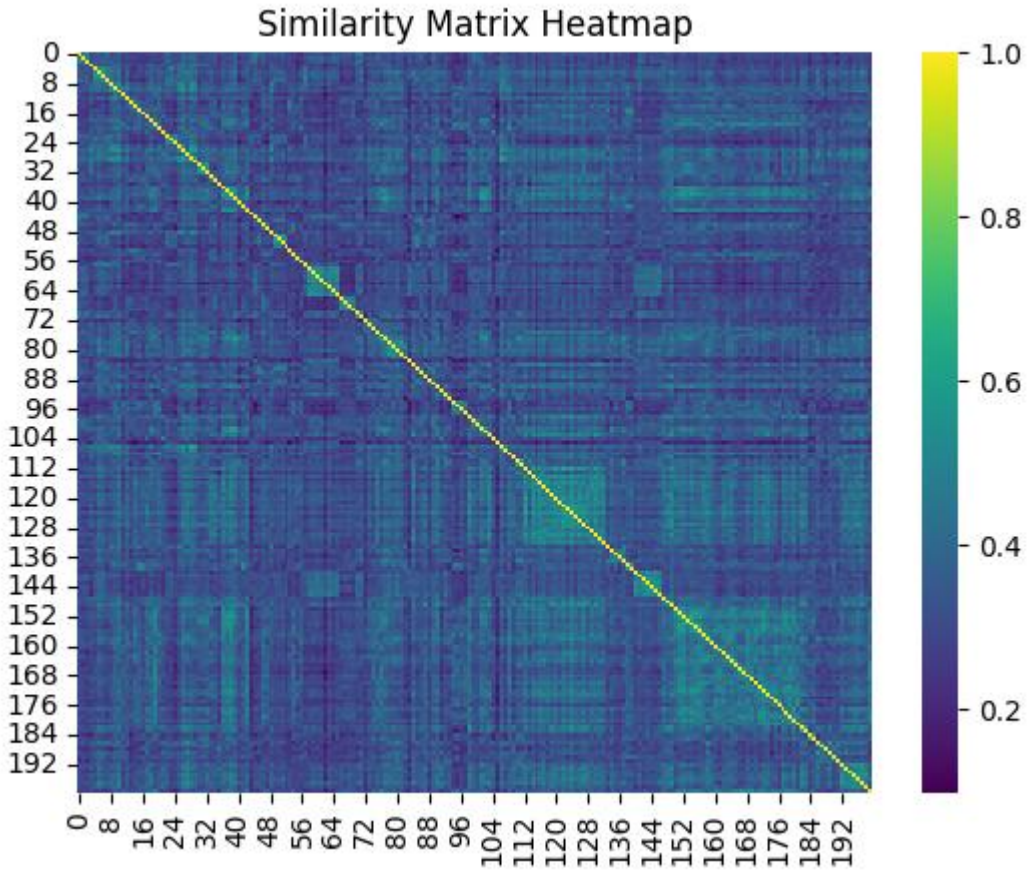
Part 1: linear probing

Your **Job 5**:

- Based on the learned $\{\boldsymbol{w}_c\}_{c=1}^{K}$, you are to compute a *K*-by-*K* cosine similarity matrix recording the cosine similarity of $\boldsymbol{w}_c$ and $\boldsymbol{w}_{c'}$
- Visualize the matrix
- Do you see some block structures? Can you explain why? Can you check the species names falling into some of the blocks?

Part 1: linear probing

Example results



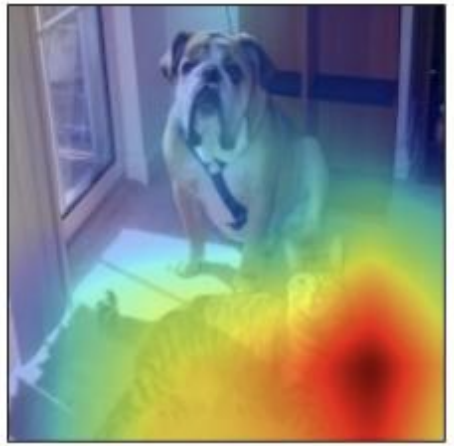
Part 1: linear probing

Your **Job 6**: repeat all steps, but switch from ResNet-50 to DINO-v2!

Part 2: Explainable AI (ResNet-50 only)

Implement Grad-CAM for visualization:

https://arxiv.org/pdf/1610.02391



(f) ResNet Grad-CAM 'Cat'

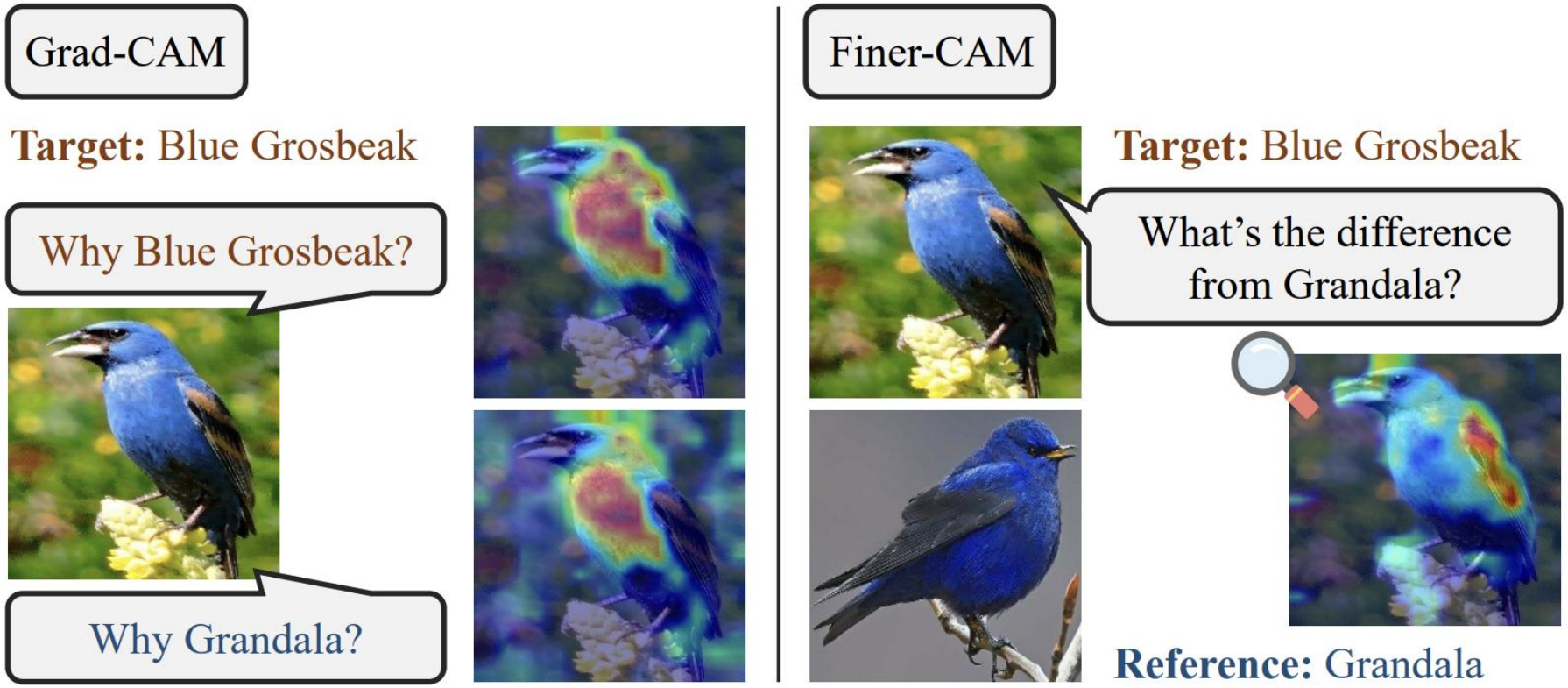


(l)ResNet Grad-CAM 'Dog'

Part 2: Explainable AI (ResNet-50 only)

Implement Finer-CAM for visualization:

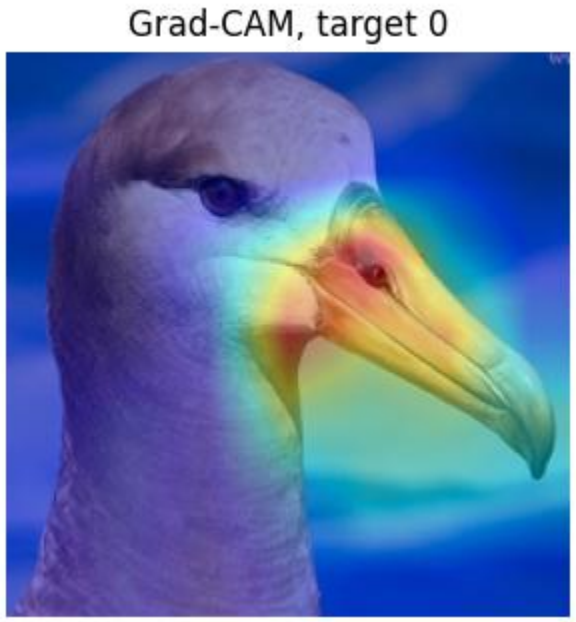
https://arxiv.org/pdf/2501.11309



Part 2: Explainable AI (ResNet-50 only)

Example results









Submission

- Your completed Google Colab ipynb file named BUusername.ipynb **(70%)**

- A report (PDF) named BUusername.pdf: **(30%)**
  - Part 1, job 4 – plots + description of observations (max 150 words)
  - Part 1, job 5 – plots + description of observations (max 150 words)
  - Part 1, job 6 – report the corresponding results of jobs 4 & 5 (max 150 words each)
  - Part 2, job 7 – some plots + description of observations (max 250 words)
  - Part 2, job 8 – some plots + description of observations (max 250 words)
  - Your AI usage disclosure: what did you use and how did you use it (max 150 words)
  - With whom did you discuss your homework?

- Each description longer than the cap will get a 30% grade reduction.